Set 34: The pH scale

The pH scale is a convenient way of expressing concentrations of hydrogen ions, $[H^+]$ in aqueous solutions. The definition of pH is expressed by the equation:

$$pH = -\log_{10}[H^+] \text{ where pH is a number without units.}$$

The pH scale has a usual range of 0 to 14 as illustrated in the diagram below:

pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$[H^+]$	$1=10^0$	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}	10^{-9}	10^{-10}	10^{-11}	10^{-12}	10^{-13}	10^{-14}
	Acidic pH < 7							Neutral pH = 7	Basic pH > 7						

Notes

Examples

1. Hydrochloric acid is one of the most common acids with many uses. It is produced in Western Australia in large quantities as a by product of the refining of titanium dioxide from the mineral sand ilmenite. Low concentration solutions of the acid can be standardised and used in analysis of basic solutions. Calculate the $[OH^-]$, $[H^+]$ and pH of a 0.0200 mol L^{-1} $HC\ell$ solution

(a) Write an ionisation equation:

$$HC\ell(aq) \rightarrow H^+(aq) + C\ell^-(aq)$$

(b) Calculate the hydrogen ion concentration:
As $HC\ell$ is a strong acid, it is fully ionised.
So the $[H^+] = 0.0200$ mol L^{-1}

(c) Calculate the hydroxide ion concentration:
Since:

$$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}$$

$$[OH^-] = \frac{1.00 \times 10^{-14}}{[H^+]} = \frac{1.00 \times 10^{-14}}{0.0200} = 5.00 \times 10^{-13} \text{ mol L}^{-1}$$

(d) Calculate the pH of the solution:

$$pH = -\log_{10}[H^+] = \log_{10}(0.0200) = 1.70$$

2. Barium hydroxide is one of the few soluble metal hydroxides. It can be used in the analysis of acidic solutions. Calculate the $[H^+]$, $[OH^-]$ and pH of a 3.00×10^{-3} mol L^{-1} $Ba(OH)_2$ solution.

(a) Write a dissociation equation:

$$Ba(OH)_2(s) \rightarrow Ba^{2+}(aq) + 2OH^-(aq)$$

(b) Calculate the hydroxide ion concentration:
As $Ba(OH)_2$ is ionic it is fully dissociated.

$$[OH^-] = 2 \times [Ba(OH)_2] = (2)\,(3.00 \times 10^{-3}) = 6.00 \times 10^{-3} \text{ mol L}^{-1}$$

(c) Calculate the hydrogen ion concentration:
Since:

$$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}$$

$$[H^+] = \frac{1.00 \times 10^{-14}}{[OH^-]} = \frac{1.00 \times 10^{-14}}{6.00 \times 10^{-3}} = 1.67 \times 10^{-12} \text{ mol L}^{-1}$$

(d) Calculate the pH of the solution:

$$pH = -\log_{10} [H^+] = \log_{10} (1.67 \times 10^{-12}) = 11.8$$

3. Orange juice has a pH of 4.50. What are the $[H^+]$ and $[OH^-]$ concentrations in orange juice?

(a) Calculate the hydrogen ion concentration:

$$pH = -\log_{10} [H^+]$$
$$-4.50 = \log_{10} [H^+]$$
$$[H^+] = 10^{-4.50} = 3.16 \times 10^{-5} \text{ mol L}^{-1}$$

(b) Calculate the hydroxide ion concentration:
Since:

$$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}$$

$$[OH^-] = \frac{1.00 \times 10^{-14}}{[H^+]} = \frac{1.00 \times 10^{-14}}{3.16 \times 10^{-5}} = 3.16 \times 10^{-10} \text{ mol L}^{-1}$$

Set 34: Exercises

In all of the following questions the temperature is assumed to be 25 °C

1. Calculate the $[H^+]$, $[OH^-]$ and pH of the following solutions.

(a) A 0.100 mol L^{-1} $HC\ell$ solution.
(b) A 0.00500 mol L^{-1} HNO_3 solution.
(c) A 0.0100 mol L^{-1} NaOH solution.
(d) A 2.00 mol L^{-1} $HC\ell$ solution used to dissolve limestone prior to analysis.
(e) A 2.45 mol L^{-1} $NaC\ell$ solution used to preserve olives.

Notes

Set 34: The pH scale

Notes

2. Calculate the $[H^+]$ and $[OH^-]$ of solutions with the following pH:
   (a) Lemon juice of pH 3.00
   (b) Dish washing solution of pH 11.0 made from dishwashing powder used in a dishwasher.
   (c) Pool acid of pH -1.00
   (d) Orange juice of pH 4.56
   (e) Swimming pool water of pH 7.60

3. Before disposal, a hydrochloric acid solution of pH 4.00 used to remove rust from iron parts prior to galvanising needs to be neutralised. Base is added until the pH is 7.00. By what factor has the hydrogen ion concentration changed?

4. A brick cleaner needs to replenish the hydrochloric acid solution he is using. Using a pH meter he measures the pH of his depleted solution to be 2.00. He decides to add 3.00 L of 3.00 mol L^{-1} $HC\ell$ solution to 2.00 L of his solution. Calculate the final hydrogen ion concentration of the new solution.

5. A laboratory technician wants a low concentration hydrochloric acid solution with a pH of 5.00, for calibrating a conductivity meter. What volume of water must be added to a 25.0 mL sample of hydrochloric acid solution of pH 3.60 to produce the required solution?

6. A researcher studying the extraction of alumina from bauxite requires 100 mL of a sodium hydroxide solution of pH 12.0. When the stock solution was checked it was found to have a pH of 11.7. What mass of sodium hydroxide must be added to increase the pH of this solution to 12.0? Assume the volume of the solution does not change with the addition of the solid.

Question 7: Concrete batching plant

7. An environmental chemist is conducting research into highly alkaline solutions found in a drain near a concrete batching plant. The concentration of the solution was found to be 0.236 mol L^{-1} hydroxide ion and the runoff from the plant was found to have a concentration of hydroxide ion of 0.156 mol L^{-1}. The chemist mixed 200 mL of the solution from the drain with 300 mL of the runoff from the plant.
   (a) Calculate the pH of the resulting solution.
   (b) What volume of 1.00 mol L^{-1} hydrochloric acid solution does the chemist need to add per litre of the mixed solution to neutralise it.
   For both (a) and (b) Assume the volumes are additive.

8. A water tank contains 15 000 L of bore water of pH 5.50. In an effort to increase the pH 10.0 g of sodium hydroxide was added to the water in the tank. Determine the pH of the resulting solution. Assume the volume does not change on adding the solid.

9. A swimming pool contains 2.00 ML of water at a pH of 7.80 making swimming in it unpleasant. The caretaker needs to reduce the pH to 6.80. What volume of 12.0 mol L^{-1} hydrochloric acid does he need to add? Assume the volume of acid is insignificant compared to the volume of the pool.